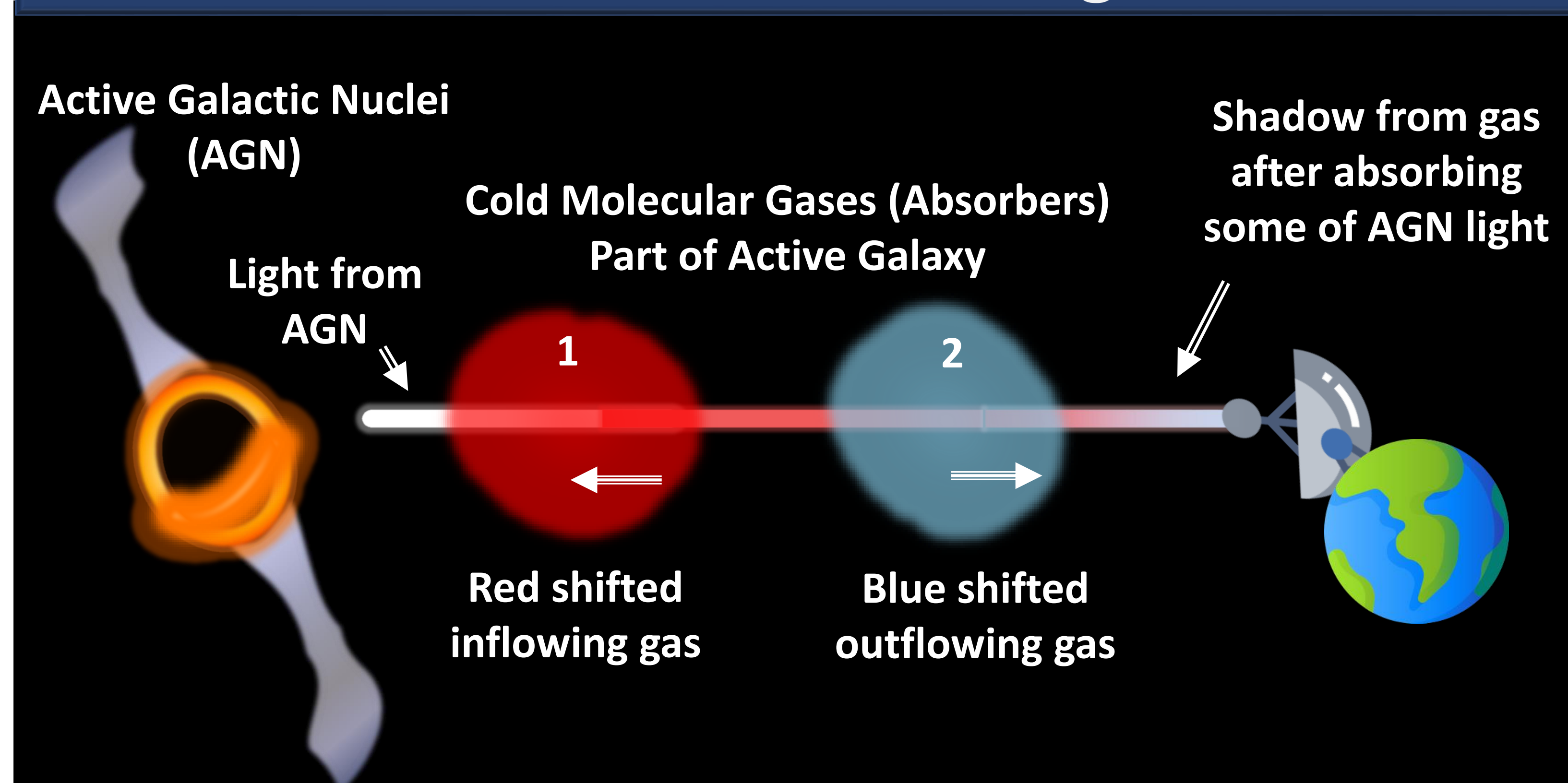
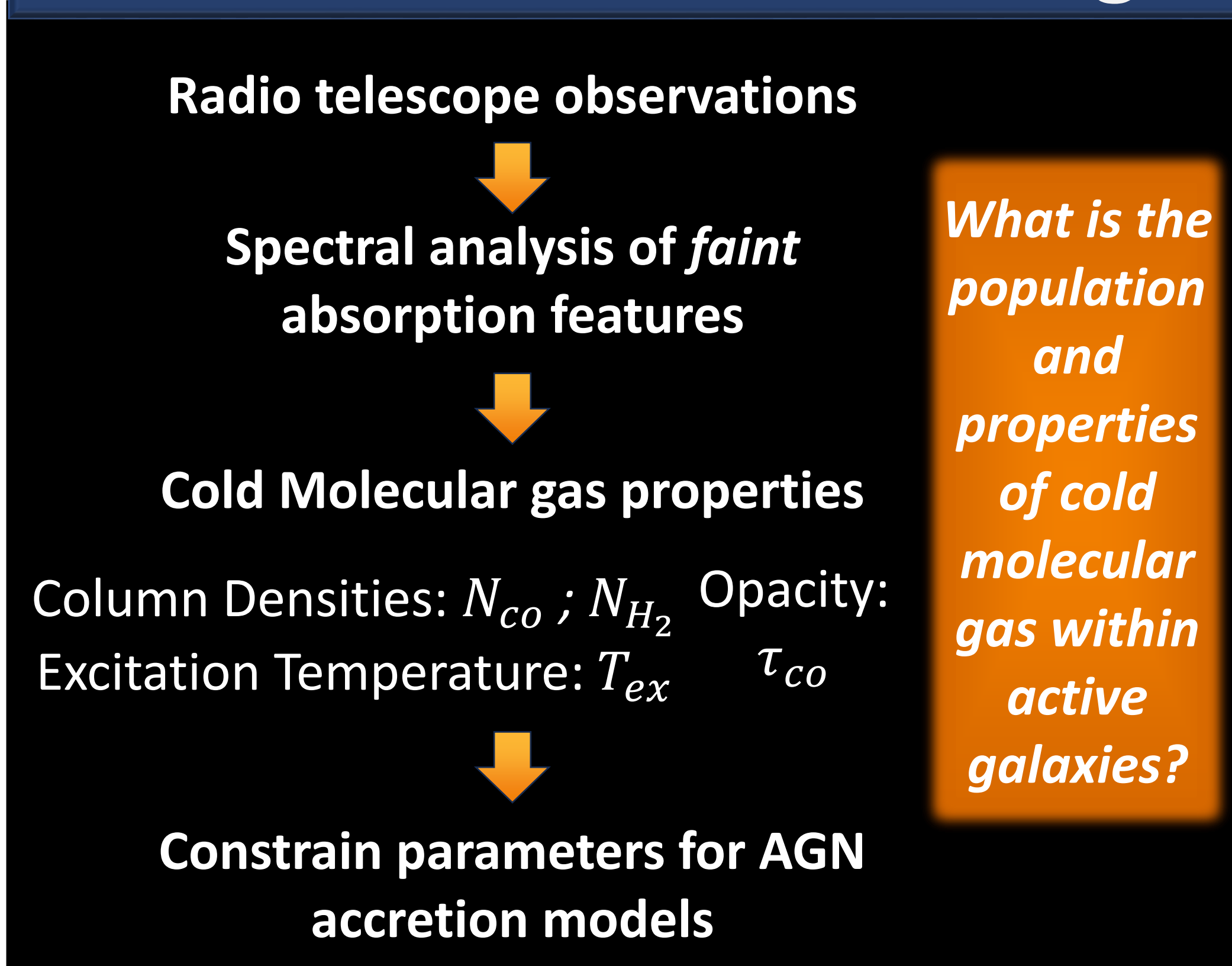


Molecular Gas Clouds Casting Shadows



Active Black Hole Feeding

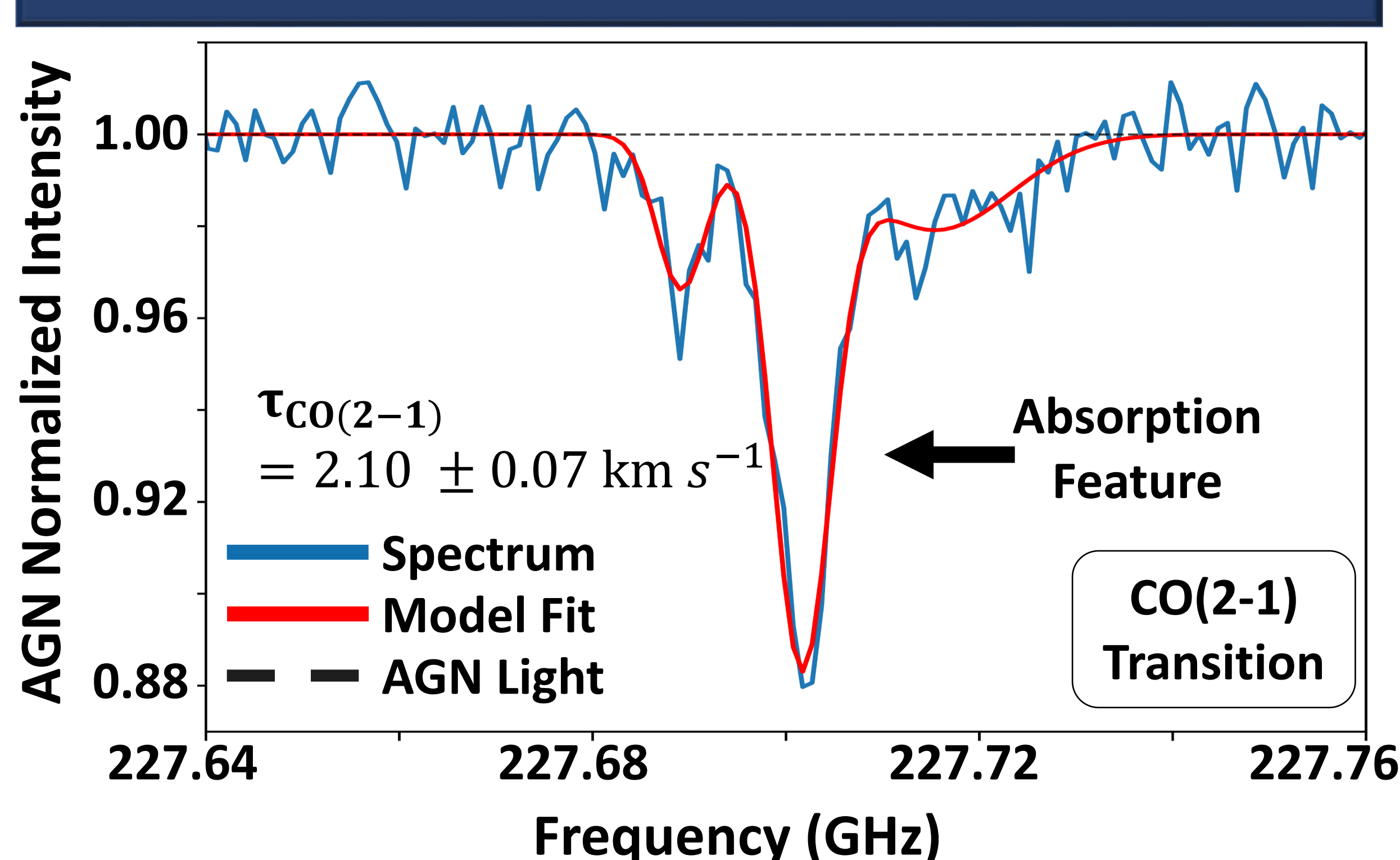


Sample Target Galaxies

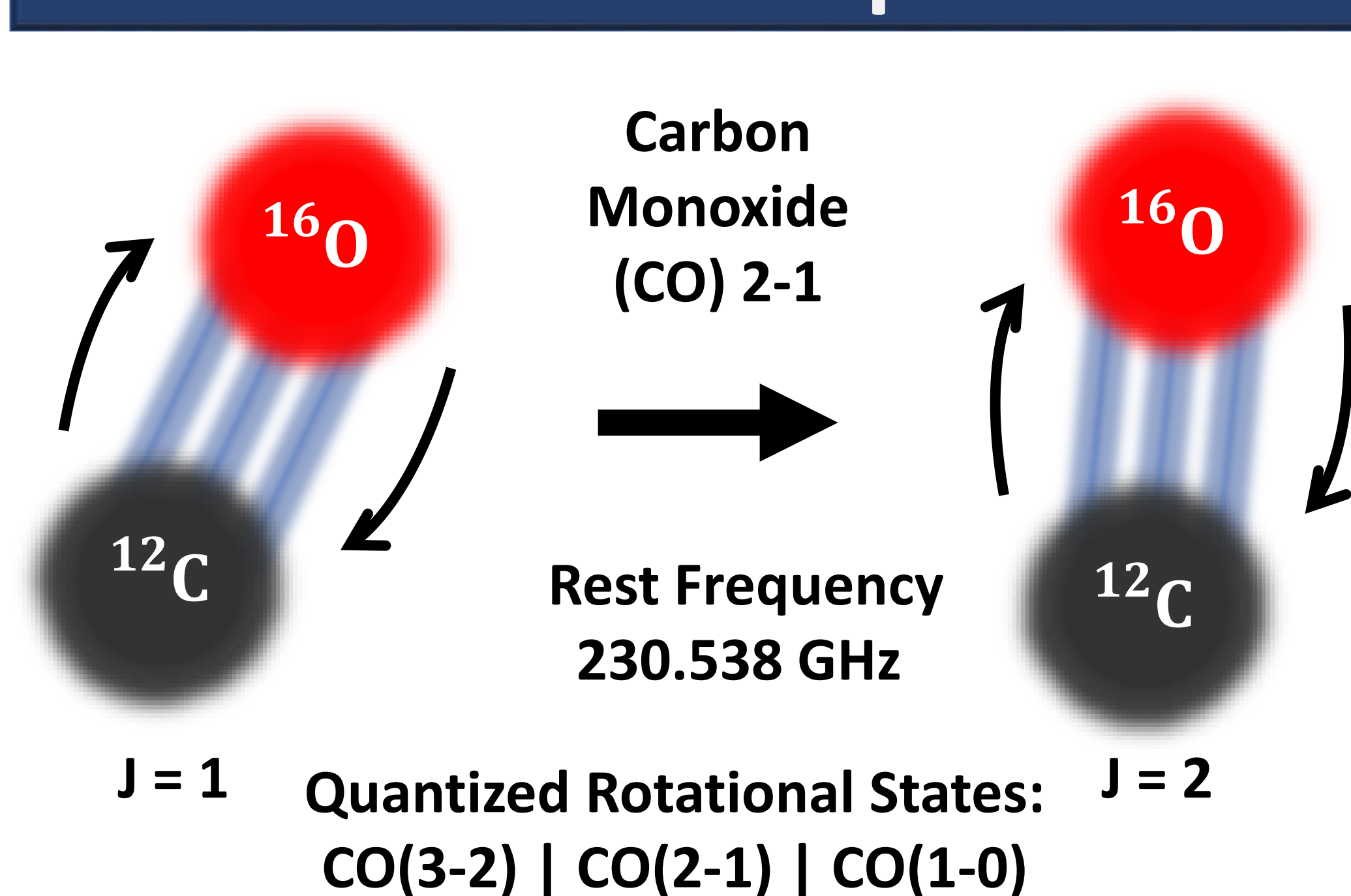


- Peak intensity: > 25 mJy
- Redshift: $z < 0.06$
- Contains CO(3-2; 2-1; 1-0)
- Observed by ALMA bands 3, 4, and 6.
- 21 total targets

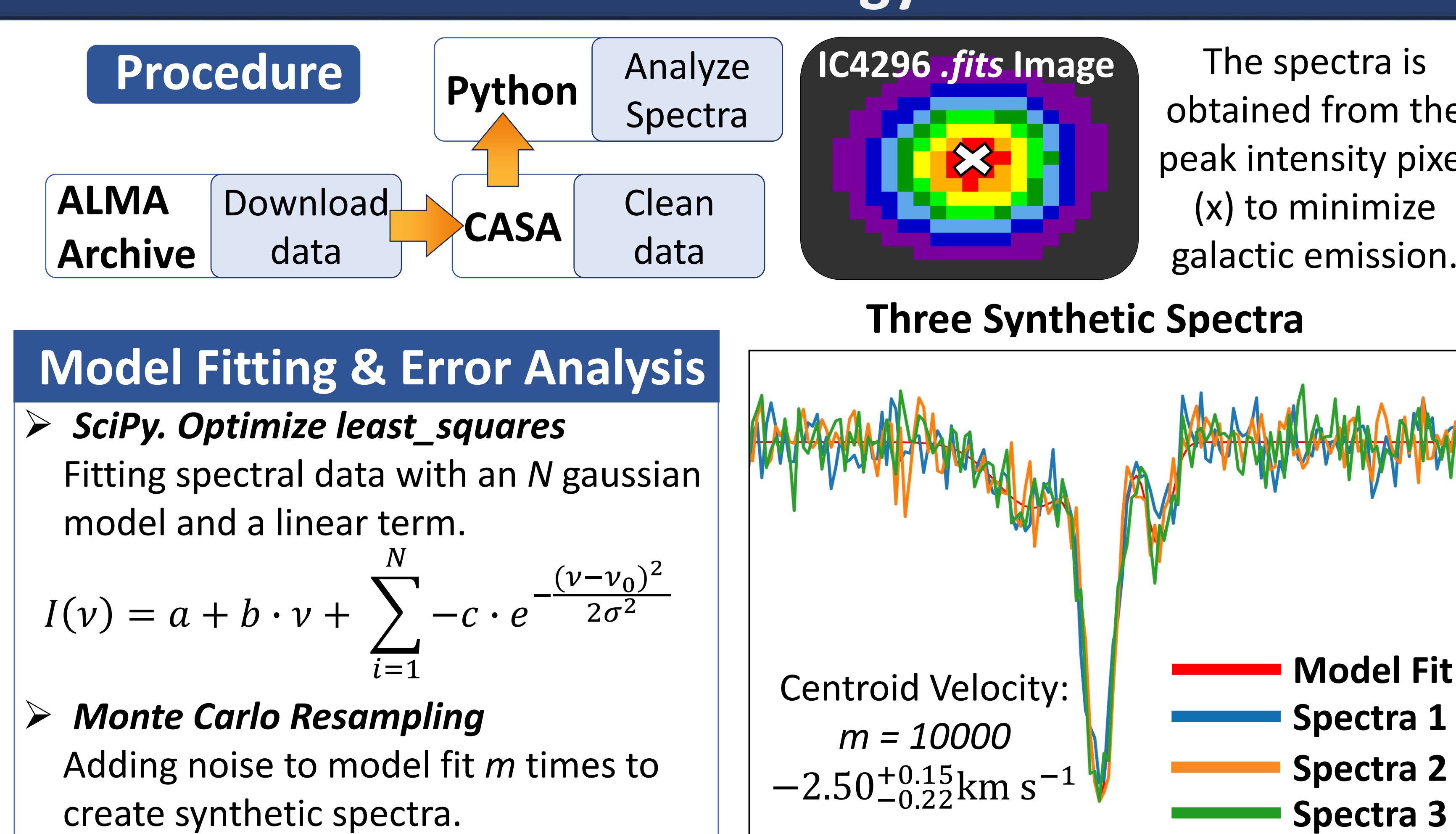
IC4296 Observed Shadow Feature



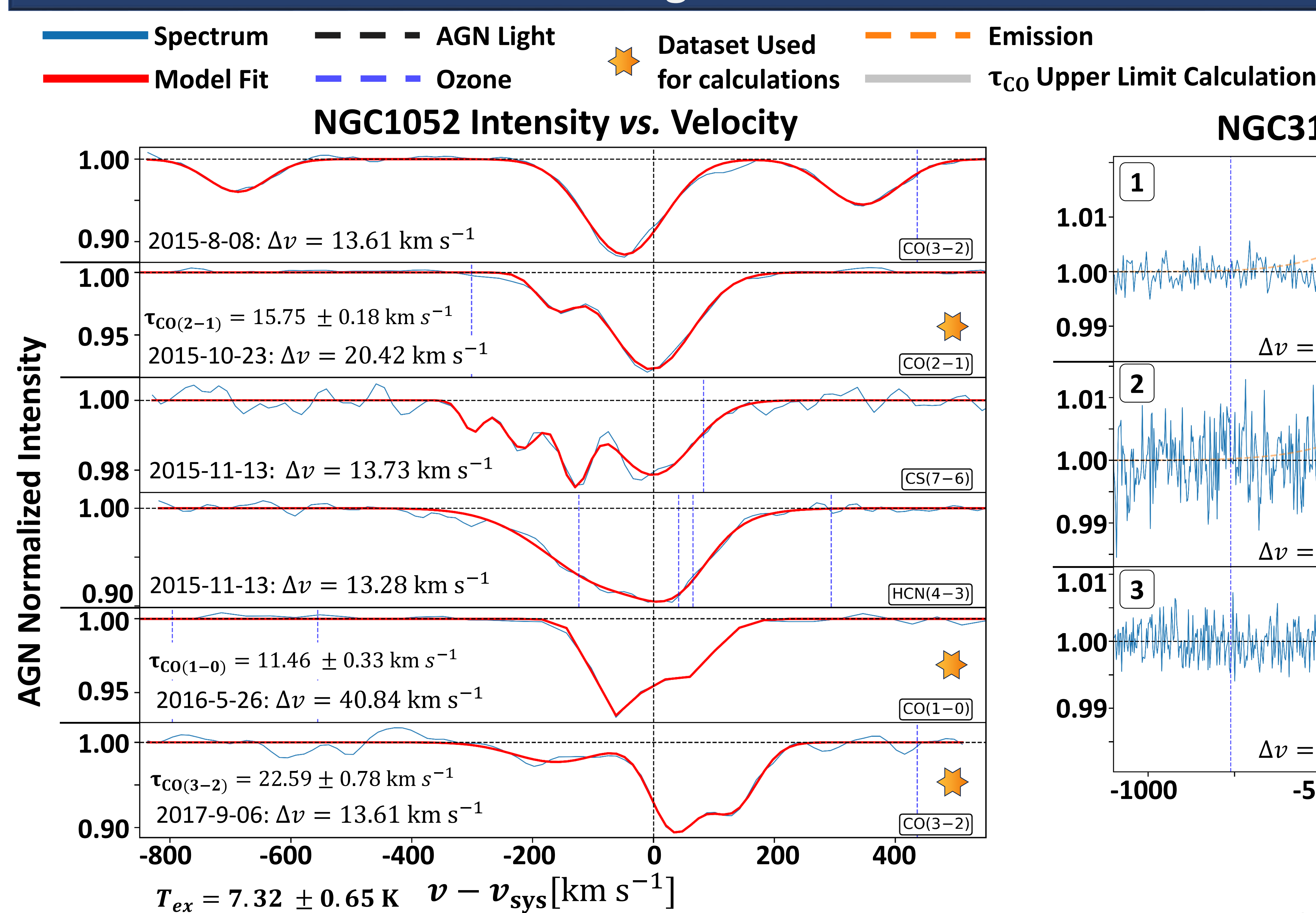
Shadow Culprit



Methodology

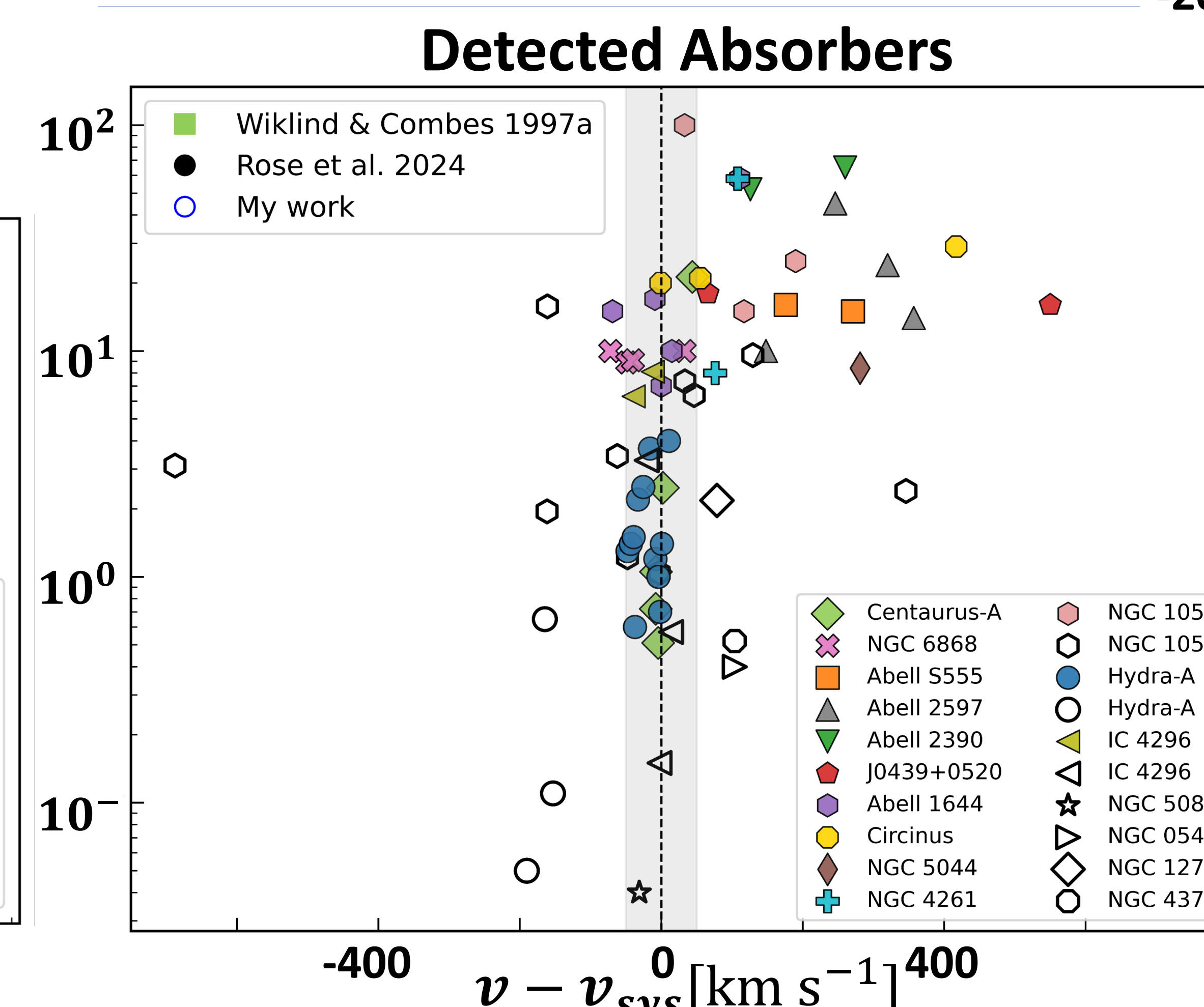
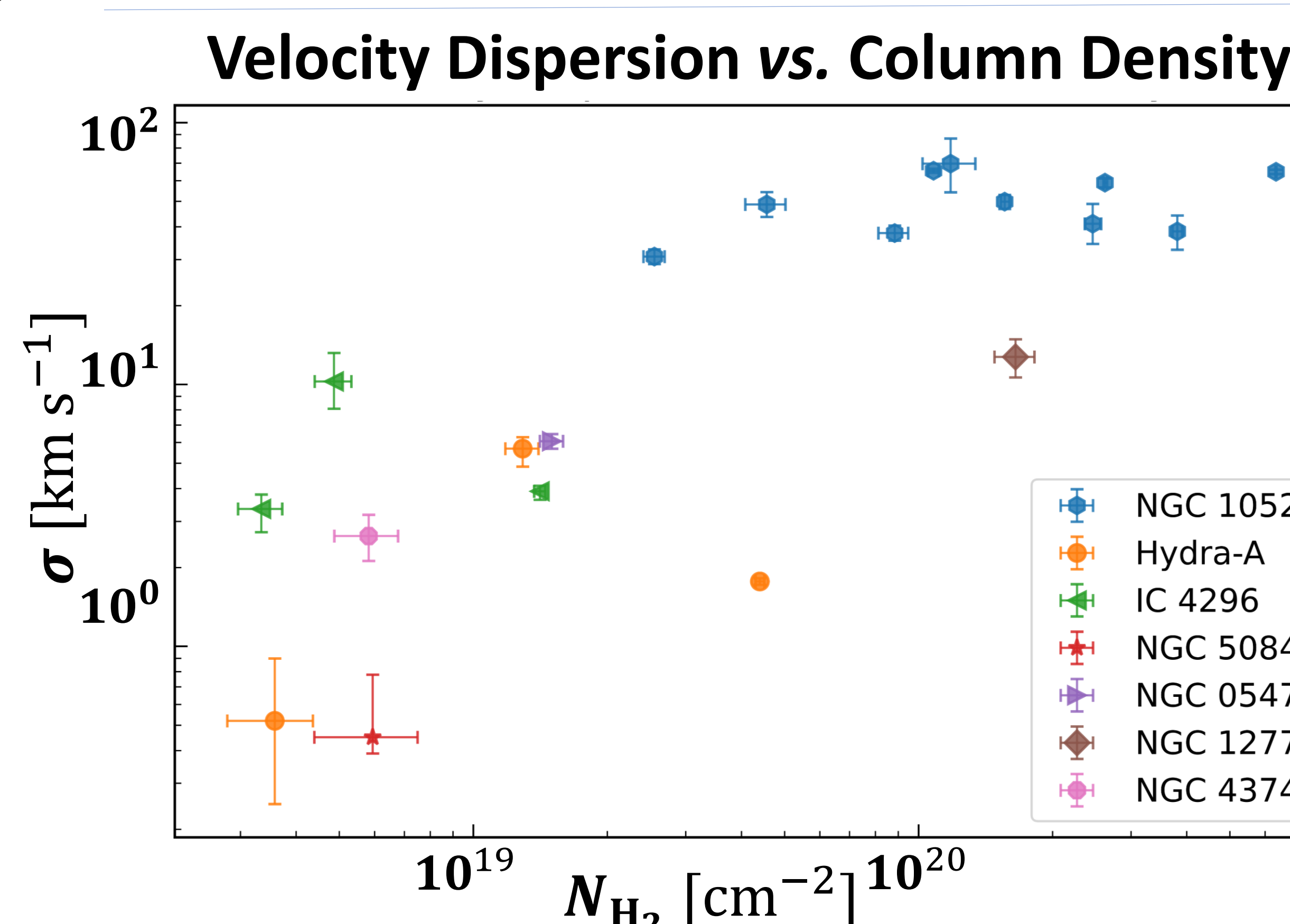


Interesting Case Studies



Results

- 7/21 targets yielded absorption features.
- The rest provide upper opacity limit calculations.
- The obtained data is currently being used to construct a mock line of sight observation simulation to model potential distributions of cold molecular gas within active galaxies.



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References

